Aspect-Based Multi-Label Sentiment Analysis and Star Rating Mismatch Identification in Google Maps Reviews of Office Buildings

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***Abstract*— In the digital era, user reviews on Google Maps have become a crucial source to assess the quality of office building services. However, star ratings often do not fully reflect the content of reviews. This study proposes an Aspect-Based Multi-Label Sentiment Analysis (ABSA) approach to detect mixed polarities and rating mismatches between user opinions and star ratings. Data were collected from Google Maps reviews and manually annotated according to five aspects: cleanliness, accessibility, service, security, and parking. An Exploratory Data Analysis (EDA) was performed to identify the distribution of aspects and sentiments, as well as to uncover dominant opinion patterns. The EDA provides strategic insights into service areas that need improvement, based on the frequency of user criticism or appreciation. A classification model based on IndoBERT was trained to automatically detect aspects and sentiments in a multi-label format. The evaluation demonstrated very high performance, achieving 99% precision, 100% recall, and 99% F1-score (micro and macro average) across three sentiment classes (positive, neutral, negative), with a total support of 605 samples. This indicates excellent accuracy in handling complex and diverse user opinions. This approach offers a data-driven method to understand and enhance office building service quality more objectively and responsively based on customer experiences.**

Keywords—ABSA, Google Maps Review, IndoBERT, Sentiment Analysis.

I. INTRODUCTION

Online review services have become a primary source of information for assessing the quality of products or locations. Platforms such as Google Maps serve not only as providers of geographic location points but also as platforms for users to share their experiences through reviews and star ratings [1]. These ratings play a significant role in influencing subsequent consumer decisions when selecting services or locations that align with their preferences and needs [2]. User-generated content, in large quantities, not only reflects individual experiences but also has the potential to shape public perceptions of business entities [3]. In this context, reviews that assess aspects such as cleanliness, accessibility, service, security, and parking facilities provide valuable information that can be leveraged to improve the management and quality of services at a location [4].

However, challenges arise in using star ratings as a comprehensive representation of user experiences [5]. Often, star ratings do not fully reflect the detailed opinions contained in reviews and can even cause mismatches between the rating score and the actual content of the review [6]. Traditional sentiment analysis approaches, which only classify sentiment in general terms toward an entity, are insufficient, as a review may encompass multiple aspects simultaneously [7], [8].

To address this issue, the Aspect-Based Sentiment Analysis (ABSA) approach, which can identify, and separate sentiment based on specific aspects within a review, becomes crucial [9]. This method, particularly when based on transformer models such as BERT [10], has been proven to provide high precision in natural language processing and aspect-based sentiment analysis [11]. This study employs the IndoBERT model, a version of BERT adapted specifically for the Indonesian language [12], to identify specific aspects and detect mismatches between reviews and ratings in office buildings [13]. The results of the analysis are expected to provide in-depth strategic insights for building management to improve service quality and customer satisfaction. This study also aims to answers and explanations related to two research questions on ABSA sentiment analysis to identify rating mismatches in Google Maps reviews:

1. How can the Aspect-Based Sentiment Analysis (ABSA) approach address the issue by separating and identifying sentiment based on specific aspects within reviews?
2. What is the effectiveness of using the IndoBERT model in analyzing aspect-based sentiment in Google Maps reviews for office buildings, and how can this model detect mismatches between star ratings and review content?

This research is organized into several interrelated sections. Chapter 2 discusses previous studies that examine the comparison of application technologies and methods used to measure the performance of these applications. Chapter 3 provides a detailed explanation of the data set used, along with relevant characteristics and the research methodology applied. Chapter 4 presents the results of the analysis and the main findings obtained from this study. Chapter 5 concludes the research findings and reflects on the insights gathered. Within this framework, the study contributes (i) a manually annotated multi-aspect dataset of Google Maps office building reviews, (ii) a fine-tuned IndoBERT model for aspect-based multi-label sentiment analysis, and (iii) a framework for detecting mismatches between textual sentiment and star ratings.

II. RELATED WORK

The findings from the three studies below reinforce the understanding of the effectiveness of using the Indonesian

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language-based BERT model in aspect-based sentiment analysis, which can be integrated and compared with the approach applied in this research.

A. Sentiment Analysis of Google Maps Reviews

This study explores the role of Google Maps reviews as a key factor in consumer decision-making, particularly in restaurants. By collecting 5,427 reviews, text mining and sentiment analysis methods were employed to map keywords that represent positive and negative perceptions. Using Random Forest and TF-IDF algorithms, the study developed a sentiment lexicon based on four criteria: food, price, service, and atmosphere. The findings provide an in-depth view of consumer preferences and can be used as a foundation for restaurant marketing strategies. In addition to emphasizing the capability of machine learning in analyzing review data, the study highlights the dominance of negative reviews, especially regarding service, reflecting consumer sensitivity to poor experiences. Despite limitations such as subjective biases in ratings and manual word classification, the results of this study open opportunities for future research with larger datasets and more varied algorithms [14].

B. The Use of BERT in Sentiment Analysis

This study discusses the application of aspect-based BERT models for sentiment analysis on open-ended responses in an Arabic-language educational survey. Utilizing a dataset from King Abdulaziz University, the study evaluates the performance of various BERT models, including FAST-LCF-ATEPC, AraBERT, MARBERT, and QARiB, in tasks such as aspect extraction, polarity classification, category mapping, and polarity aggregation. The results indicate that FAST-LCF-ATEPC (multilingual) excels in aspect extraction with an F1-score of 0.58, while AraBERT provides broader coverage, although with lower precision. These models build an end-to-end system that automatically maps, evaluates, and summarizes student opinions based on specific aspects in surveys, thus facilitating decision-making for educational institutions. This study fills the gap in ABSA research for Arabic text in the educational sector, an area that has been less explored. The success of the FAST-LCF-ATEPC model in detecting polarity and aspects, supported by an aspect mapping accuracy of 98%, demonstrates the significant potential of transfer learning in natural language processing [15].

C. Aspect Based Sentiment Analysis Using IndoBERT

This study evaluates three IndoBERT-based BERT models for aspect-based sentiment analysis (ABSA) on customer reviews in Indonesia. The first model combines IndoBERT, CNN, and machine learning (XGBoost, Random Forest, Naïve Bayes) for single-sentence classification. The second model uses fine-tuned IndoBERT for single-sentence classification, while the third model employs fine-tuning IndoBERT with sentence pairs (review sentence and auxiliary sentence). The findings indicate that the third model with NLI-B achieved the highest F1-score, outperforming baselines such as Word2Vec-CNN-XGBoost and mBERT-aux-NLI-B with an improvement of up to 23.6%. Although the third model is the most effective, it requires longer processing time, making the second model the most balanced in terms of effectiveness and efficiency. This study emphasizes the potential of IndoBERT in understanding semantic relationships in the Indonesian language to support the development of online review analysis systems [16].

In contrast to previous studies that examined restaurant reviews [14], Arabic-language survey responses [15], or general Indonesian customer reviews [16], this research focuses on office building reviews and leverages IndoBERT within a multi-label framework to capture aspect-level sentiments while also detecting sentiment–rating mismatches. By situating the study within the broader context of Google Maps research [1]–[4] and recent advances in BERT-based ABSA approaches [9]–[13], the work underscores its novelty and contribution to both domain-specific applications.

III. RESEARCH METHODOLOGY

The sequence of methods used in this study were detailed in this chapter and visualized in Fig. 1.

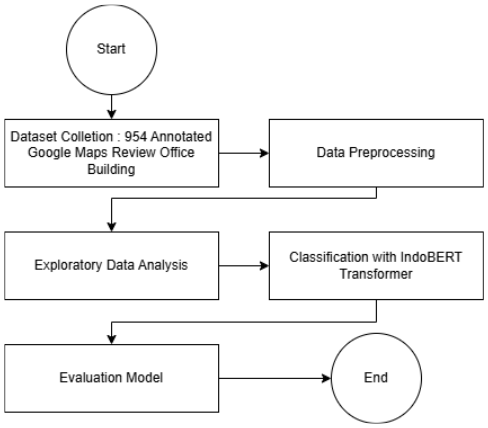


Fig. 1. Research Methodology

A. Abbreviations and Acronyms

The review data used in this study came from Google Maps Reviews of various office buildings in the Jakarta area. Each review was manually annotated to identify one or more aspects discussed and the sentiment polarity associated with those aspects, i.e., positive, negative, or neutral. The raw data used consisted of approximately 954 reviews in TABLE I.

TABLE I. GOOGLE MAPS REVIEW ANNOTATED DATA

Column Name	Count	Data Type	Description
Stars	954	int64	The numeric star rating given by users, representing overall satisfaction.
Text	419	object	The raw review text submitted by users.
Aspect_Real_1 - 5	330	object	Manually annotated aspect identified in the review.
	99		
	42		
	9		
	9		
Sentiment_Aspect_Real_1 - 5	326	object	Sentiment (positive, neutral, or negative) associated with Aspect_real
	96		
	41		
	9		
	3		

B. Data Collection and Preprocessing

In the first stage, review data were collected through scraping from the Apify platform, which accesses Google Maps reviews related to office buildings. The dataset gathered contained reviews in Indonesian that provide information on various aspects of building services. These aspects include cleanliness, accessibility, service, security,

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and parking facilities, and were annotated according to the ISO 22483:2020 standard, which provides guidelines for classifying service quality in the hospitality and office sectors in TABLE II & TABLE III which are presented in English and Bahasa.

TABLE II. SAMPLE REVIEWS MULTI ASPECTS AND SENTIMENT USING ENGLISH

Stars	Text	Aspect 1	Sentiment 1	Aspect 2	Sentiment 2
5	The concrete asphalt is already quite good, firm, and the price is friendly	Price	Positive	Product Quality	Positive
1	Their rules discriminate against the poor... they treat them poorly here	Service	Negative	Staff	Negative
4	Their rules discriminate against the poor... they treat them poorly here	Service	Negative	Staff	Negative

The data preprocessing involved manual annotation of the dataset, where each review is annotated with five relevant aspect labels.

TABLE III. SAMPLE REVIEWS MULTI ASPECTS AND SENTIMENT USING BAHASA

Stars	Text	Aspect 1	Sentiment 1	Aspect 2	Sentiment 2
5	Gedung yg nyaman dan safety coock buat meetiing dsini dan posisi pintu tol sgt dekat	Access	Positive	Comfort	Positive
5	Gedungnya bagus, rapi, dan bersih. pelayanannya sangat memuaskan dan fasilitasnya sangat memadai	Facility	Positive	Service	Positive
5	Gedungnya bagus dan dari parkirannya bisa naik langsung sampai lt 6	Parking	Positive	Building	Positive

Each review was also analyzed to determine the sentiment polarity toward these aspects, i.e., whether the user's opinion is positive, negative, or neutral. This annotation process ensured that the dataset is ready for more in-depth and accurate aspect-based sentiment analysis, specifically in identifying mismatches between star ratings and the actual content of the reviews (rating mismatch).

C. *IndoBERT Transformer*

After preprocessing and data exploration, the core stage of this research was the training of a multi-label classification model using the IndoBERT architecture. IndoBERT was selected because it is a pre-trained version of BERT tailored to the Indonesian corpus, making it more accurate in handling local structures and semantics. The model was developed with a fine-tuning approach, where the parameters of the pre-trained IndoBERT model are adapted to the manually annotated office building review dataset. The dataset contains input pairs in the form of review text and multi-labels (comprising aspects and sentiment). The training process was carried out with the following configuration in TABLE IV.

TABLE IV. INDOBERT TRANSFORMER CONFIGURATION

Component	*Configuration*	*Description*
Pretrained Model	indobenchmark/indobert-base-p1	A pre-trained IndoBERT model provided by IndoNLU, trained on a large Indonesian corpus and used as the base for the multi-label classification task.
Tokenization	BertTokenizer from Huggingface Transformers	Responsible for converting review texts into token sequences that are compatible with the BERT architecture while maintaining the semantic structure of the input.
Task	Multi-label Classification	Since each review may be associated with more than one aspect and sentiment, a multi-label classification approach was used instead of single label.
Loss Function	Binary Cross Entropy with Logits	This loss function treats each label as an independent binary classification problem, making it suitable for multi-label settings where multiple labels may be true per instance.
Optimizer	AdamW	A popular optimizer for fine-tuning transformer models, known for effective weight regularization and handling of sparse gradients.
Epoch	3	The model was trained over 3 full passes of the dataset, which was found sufficient for convergence without overfitting.

The model output was a probability vector per label, which is then binarized using a threshold of 0.5 to produce the final predictions. Evaluation was performed by calculating classification metrics such as precision, recall, and F1-score (micro, macro, weighted), and was displayed in the form of a classification report and confusion matrix. The training results showed that the model successfully recognizes various combinations of aspects and sentiment polarities in a single review with high accuracy. This fine-tuning approach allowed for deep adaptation to the characteristics of Indonesian language reviews and demonstrated high performance in aspect-based multi-label classification tasks.

D. *Exploratory Data Analysis*

The Exploratory Data Analysis (EDA) stage in this study was conducted to understand the distribution and patterns of the collected review data [16]. It also helped identify data imbalances that may affect the aspect-based sentiment analysis process and provided the basis for training the BERT-based model. This analysis included visualizing the frequency distribution of reviews based on the aspects described below. Additionally, an observation of the proportion of sentiment (positive, negative, neutral) for each aspect was performed to identify dominant user opinion trends. Graphs, scatter plots, and distribution analyses are used for these observations.



Fig. 2. Word Cloud of Most Frequent Words

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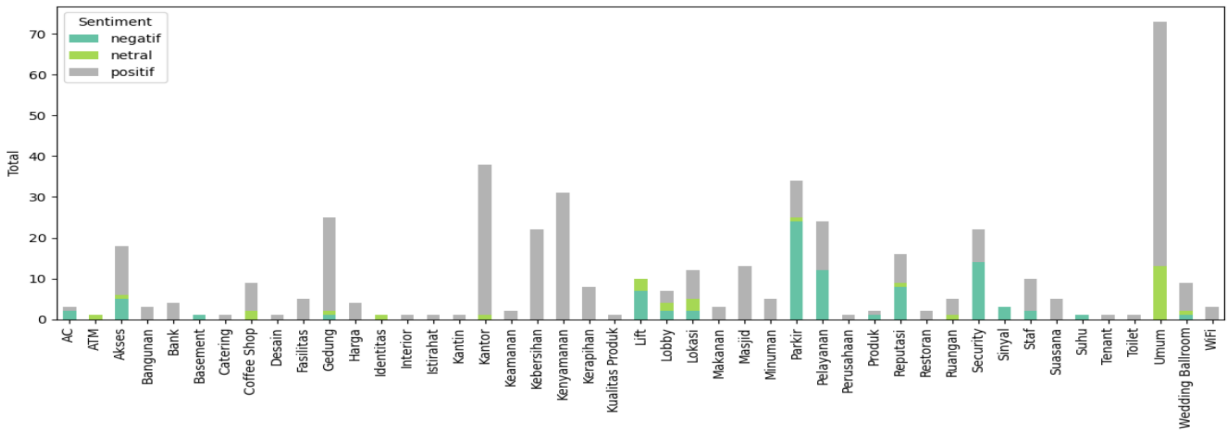


Fig. 3. Distribution of Sentiment

The word cloud visualizes the most frequent words in the cleaned_text column of Google Maps reviews, with larger words indicating higher frequency in Fig. 2. Key terms such as "kantor" (office), "bagus" (good), "nyaman" (comfortable), "parkir" (parking), and "gedung" (building) highlight users' focus on office comfort, parking facilities, and positive perceptions of the workplace environment. Words like "security", "bersih" (clean), "luas" (spacious), and "strategis" (strategic) reflect concerns about security, cleanliness, and building location. Negative terms like "kurang" (lacking) and "susah" (difficult) suggest critical reviews. Overall, the word cloud offers an initial view of user perceptions, laying the groundwork for further aspect-based sentiment analysis (ABSA).

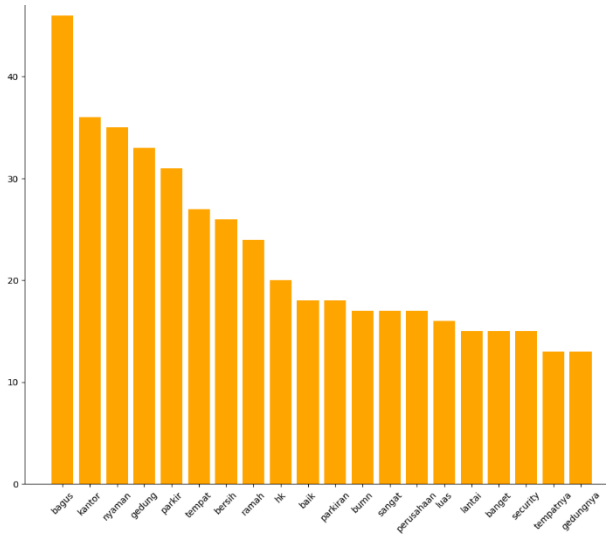


Fig. 4. Bar Chart of Most Frequent Words

The bar chart in Fig. 4 displays the 20 most frequent words in the cleaned_text column after text preprocessing. "Bagus" (good) is the most common, indicating positive sentiment, while "kantor" (office), "nyaman" (comfortable), and "gedung" (building) reflect the focus on workspace environment and comfort. Words like "parkir" (parking), "bersih" (clean), and "security" highlight concerns with facilities, cleanliness, and safety. Terms like "hk", "bumn", and "perusahaan" refer to institutional identities, and "luas" (spacious) and "lantai" (floor) emphasize physical aspects of the building. This chart provides a preliminary overview for Aspect-Based Sentiment Analysis (ABSA).

The bar chart in Fig. 3 shows the distribution of sentiment polarities (positive, neutral, negative) for each service aspect mentioned in Google Maps reviews. Each bar represents a specific aspect (e.g., parking, security, office), with colors indicating sentiment: gray for positive, light green for neutral, and turquoise for negative. Key findings include the dominance of positive sentiment for "General," indicating overall favorable reviews, and a more varied distribution for "Parking," with significant negative sentiment. Aspects like "Mosque," "Office," and "Service" are mostly positive, while "Security" and "Reputation" show notable negative sentiment. Minor aspects like "AC," "Toilet," and "Wi-Fi" have fewer opinions but still reveal useful sentiment diversity. This distribution provides strategic insights for building managers to prioritize service improvements.

The heatmap in Fig. 5 shows the sentiment distribution (positive, neutral, negative) for each service aspect identified in the user reviews. The vertical axis represents aspects such as Service, Security, Toilet, Food, Products, and others, while the horizontal axis shows sentiment polarities. Darker colors indicate higher frequencies of specific sentiment-aspect combinations. Key findings include "Wedding Ballroom" and "Comfort" having the most positive sentiment, with 60 and 31 reviews, respectively, showing high user satisfaction. "Products" and "Security" have the most negative sentiments, with 24 and 14 reviews. "Service" has a balanced distribution of sentiments, while aspects like "AC," "Coffee Shop," "Interior," and "Wi-Fi" are rarely mentioned or have neutral sentiment, suggesting lower user focus.

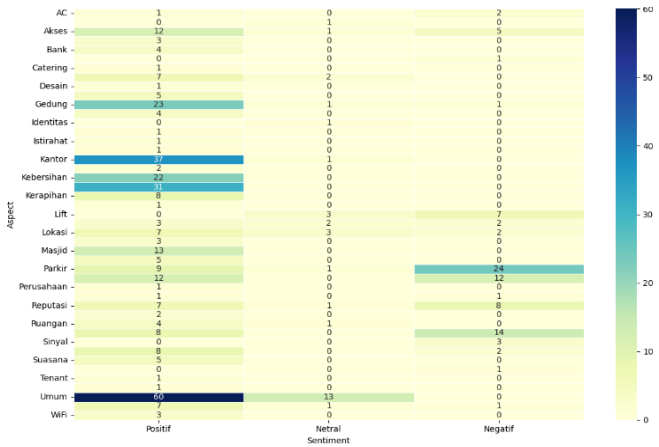


Fig. 5. Heatmap of Each Aspect's Sentiment Distribution

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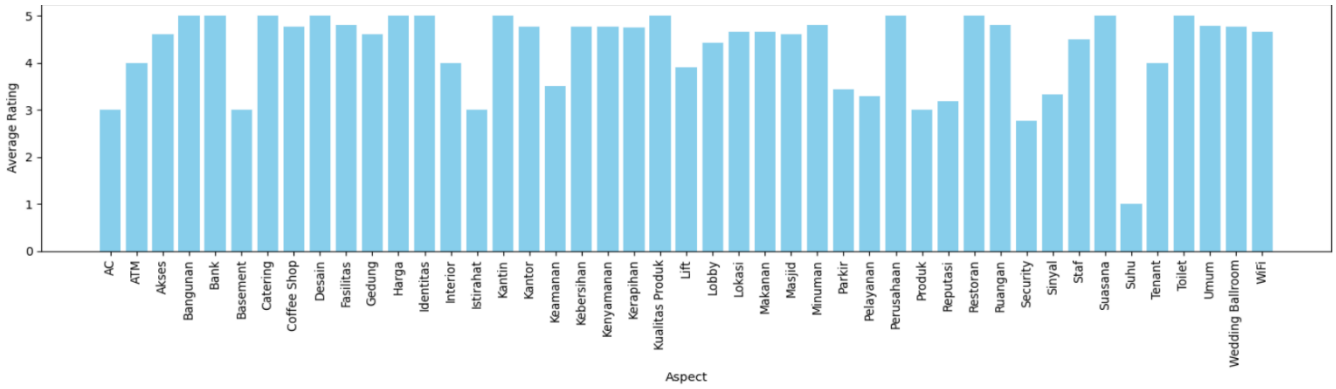


Fig. 6. Bar Chart of Each Aspect's Average Rating

The bar chart in Fig. 6 displays the average star rating (1 – 5 scale) given by users for each office building service aspect. Most aspects have an average rating above 4.5, indicating high user satisfaction, particularly for AC, Facilities, Office, Wedding Ballroom, and Accessibility. Aspects like Mosque, Reputation, Parking, and Company have lower average ratings (3.0-3.5), suggesting dissatisfaction or unmet expectations. Security and Toilet have the lowest ratings, nearing 1 and 2, signaling strong negative perceptions and the need for priority attention. Meanwhile, Wedding and Ballroom aspects received a perfect rating (5.0), reflecting highly positive user experiences and serving as key selling points. The results of the EDA are crucial in validating the completeness and representativeness of the dataset before it is applied to the ABSA model using IndoBERT.

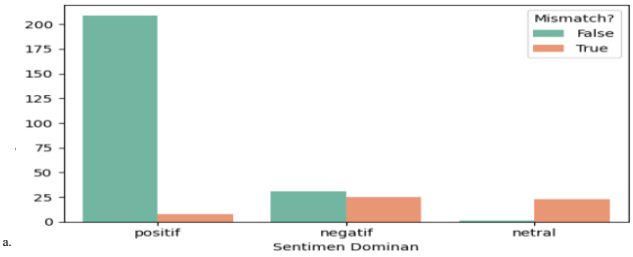
IV. RESULT & DISCUSSION

In this study, an Aspect-Based Sentiment Analysis (ABSA) approach was applied to analyze Google Maps user reviews on office building services. ABSA functions to simultaneously identified two main elements: the Topic or service aspect and the sentiment polarity. By utilizing this approach, the study can uncover more specific opinions, such as in a review stating "security is poor but parking access is very convenient." The model was capable of capturing dual sentiments within a single review, something that cannot be achieved by general sentiment analysis. Evaluation is conducted using Precision, Recall, and F1-Score metrics, with data support or ground truth available for each class in the test dataset. The model results are presented in TABLE V, which summarizes the performance of the model for each sentiment label.

TABLE V. SAMPLE REVIEWS MULTI ASPECTS AND SENTIMENT USING BAHASA

Sentiment	*Precision*	*Recall*	*F1-Score*	*Support*
Negatif	99%	100%	100%	169
Netral	99%	100%	99%	194
Positif	99%	99%	99%	242
Micro avg	99%	100%	99%	605
Macro avg	99%	100%	99%	605

In total, there were 605 reviews in the test data used as the basis for evaluation (TABLE V). The micro F1-score of 99% indicates that the model has an exceptionally high ability to accurately identify combinations of aspects or topics and sentiments. The high precision and recall also suggest that the model's predictions are not only accurate but also comprehensive in capturing user opinions. With consistent performance across all three sentiment classes (negative, neutral, and positive), this model is highly reliable in understanding public reviews in depth. The combination of the ABSA approach and IndoBERT has proven effective in classifying service topics and user opinion polarities simultaneously, providing strategic insights for data-driven decision-making in service quality improvement.



a.

Fig. 7. Sentiment Dominant vs Missmatch Rating

Most reviews align between textual sentiment and star ratings, as reflected by the predominance of **"False"** in the mismatch distribution (Fig. 7). A smaller subset is flagged as mismatched (**"True"**)—for example, when a 4–5-star rating accompanies neutral or negative sentiment—likely due to politeness bias or reluctance to post negative public ratings. An illustrative case is the review "agak merepotkan", which received 3 stars but is classified as predominantly negative. These mismatches surface meaningful anomalies in user perception where sentiment and ratings diverge. Furthermore, most reviews have has_mixed_polarity = False, indicating a single prevailing sentiment. Entries with has_mixed_polarity = True typically discuss multiple aspects with divergent opinions and therefore demand careful interpretation, particularly for service-quality assessment and automated review summarization.

The heatmap in Fig. 8 summarizes the distribution of average user ratings across various topic-aspect combinations, highlighting specific service areas requiring attention. In this analysis, two combinations have been flagged for further improvement. The most critical issue is identified in the "Comfort & Environmental Conditions" category under the aspect "Temperature (Suhu)" (Rating: 1.0), which strongly suggests significant dissatisfaction related to air conditioning, ventilation, or indoor climate control. This represents a high-priority operational issue due to its direct impact on occupant comfort. Additionally, an early warning signal is identified under "Reputation & Branding" with the aspect "Security" (Rating: 2.9). This rating suggests emerging concerns related to building safety, security personnel behavior, or visitor

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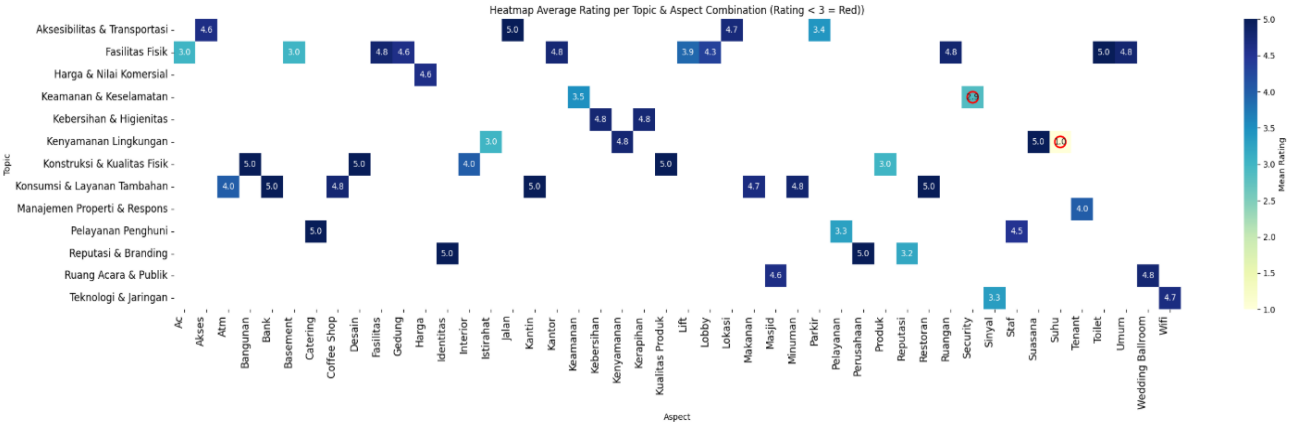


Fig. 8. Heatmap of Each Combination's Average Ratings

perception of security standards, which may eventually affect the building's brand reputation if not addressed proactively. In contrast, most combinations demonstrate excellent service delivery, with perfect ratings (5.0) observed in several areas, including Construction & Physical Quality, Event & Public Spaces, and Tenant Services. These strong-performing dimensions serve as internal benchmarks and good practices for sustaining operational excellence. Overall, while the building's service performance remains strong in many aspects, the flagged combinations clearly highlight where targeted corrective actions, monitoring, and service quality interventions should be prioritized to ensure long-term customer satisfaction and protect brand reputation.

The discussion highlights that the model's strength lay in the fine-tuning of IndoBERT on the Indonesian language dataset. The use of multi-label classification enables the model to handle multiple aspects and sentiments per review, reflecting the real-world nature of user reviews.

V. Conclusion/Future Work

This study successfully developed an Aspect-Based Sentiment Analysis (ABSA) system based on IndoBERT to detect aspects and sentiment polarities in Google Maps reviews of office buildings. The system also flags mismatches between review content and star ratings. Trained on manually annotated data, it handles multi-aspect and dual-sentiment cases and achieved a micro F1-score of 0.99. A key limitation is the small post-cleaning dataset (≈300 valid reviews). Future work should expand and diversify the reviews across locations and sectors to improve generalization and robustness.

Acknowledgment

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